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MAT-225-16805-M01 Calc I: Single-Variable Calc 2025 C-6 (Oct - Dec) : MAT-225 : 2079955,
Aleksey Ahmann, 12/9/25 at 4:37:40 PM EST

Question1: Score 0.5/1

Crossing the Room

You receive a response from your home planet with details on how to escape the planet. You are directed to connect your computer to the alien computer as a means to gather information. You are told that there is a spacecraft that can get you home, but it is guarded by several gates and traps. Your computer has informed you that the raw ingredients for fuel are available, but the amount of fuel in the ship is unknown. The first step is to travel across a room that includes hidden traps in the floor. There are safe places where you must stop in order to make it across the room safely. The location of these safe spaces will be translated into mathematics, which you can solve, and then program into your computer. Your computer will then detail precisely where you can stop in the room. Your computer translates the following:

Find the open intervals on which the function $f(x) = x + 10\sqrt{6 - x}$ is increasing or decreasing. The safe points will be calculated from these intervals. If the function is never increasing or decreasing, provide an input of NA to your computer.

Increasing Interval:

Your response	Correct response
$(-\infty, -19)$	$(-\infty, -19)$

Auto graded Grade: 1/1.0 

Decreasing
Interval:

Your response	Correct response
$(-19, 6]$	$(-19, 6)$

Auto graded Grade: 0/1.0 

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

The following expression is given:

$$f(x) = x + 10 \cdot \sqrt{6-x}$$

The task at hand is to work out intervals where the slope of the function is increasing and decreasing. I did this by first working out the domain of the function, that is an interval $\{x_i \leq x \leq x_f\}$ listing valid inputs that $f(x)$ can take in and output a result from.

Because this function includes an x in a rational function, the given x value ought not to be < 0 , or else the radical will result in an imaginary number, which is not valid for this case.

Then, I will use calculus to work out the optimal point of the function, or a c^* such that $\left. \frac{df}{dx} \right|_{x=c^*} = 0$. After that, I will use a "crude, intuitive test" to observe the behaviour of the function near the c^* from both the left-hand and the right-hand sides of the graph.

Step 1: Domain of the Function

The function must take in a real number, or a subset of the reals, and output a real number, or a subset of the reals. The given expression is $x + 10 \cdot \sqrt{6-x}$, which is undefined if the x is < 0 . I worked out a range of x values such that the given expression is ≥ 0 with the following method:

$$\mathbb{D}(f) \rightarrow \text{sol.} \left\{ \begin{array}{l} x + 10 \cdot \sqrt{6-x} \geq 0 \\ \Rightarrow \sqrt{6-x} \geq 0 \\ \Rightarrow (\sqrt{6-x})^2 \geq 0^2 \\ \Rightarrow 6-x \geq 0 \\ \Rightarrow -x \geq -6 \\ \Rightarrow x \leq 6 \\ \Rightarrow \mathbb{D}(f) = (-\infty, 6] \end{array} \right.$$

The interval of x values in which the $f(x)$ can be evaluated in a valid manner is $\mathbb{D}(f) = (-\infty, 6]$.

Step 2: $f(x)$ Optimal Value

Of course, the function's optimal value needs to be worked out, which is done with the following relation:

$$\left. \frac{df}{dx} \right|_{x=c^*} = 0$$

The logic to work out the c -value is ...

$$c^* \text{ s.t. } \left. \frac{df}{dx} \right|_{x=c^*} = 0 \rightarrow \text{sol.} \left\{ \begin{array}{l} \left. \frac{df}{dx} \right|_{x=c^*} = 0 \because \text{Problem Setup} \\ \Rightarrow \frac{d}{dx} [x + 10 \cdot \sqrt{6-x}] \because f(x) = x + 10 \cdot \sqrt{6-x} \\ = \frac{d}{dx} [x] + \frac{d}{dx} [10 \cdot \sqrt{6-x}] \because \frac{d}{dx} \sum f(x) = \sum \frac{d}{dx} f(x) \\ = \frac{d}{dx} [x] + 10 \cdot \frac{d}{dx} [\sqrt{6-x}] \because \frac{d}{dx} [c \cdot f(x)] = c \cdot \frac{df}{dx} \\ = \frac{d}{dx} [x] + 10 \cdot \frac{d}{dx} [(6-x)^{1/2}] \because \sqrt{x} = x^{1/2} \\ = \frac{d}{dx} [x] + 10 \cdot \left[\frac{1}{2} (6-x)^{(1/2)-(2/2)} \cdot \frac{d}{dx} [6-x] \right] \\ \because \frac{d[(f_{\text{nest}})^n]}{dx} = (n \cdot f_{\text{nest}}^{n-1}) \cdot \frac{df_{\text{nest}}}{dx} \\ = \frac{d}{dx} [x] + 10 \cdot \left[\frac{1}{2} (6-x)^{-1/2} \cdot -1 \right] \because \frac{d}{dx} [c-x] = -1 \\ = \frac{d}{dx} [x] - \frac{5}{\sqrt{x-6}} \because \text{Simplification of the Key Term} \\ = 1 - \frac{5}{\sqrt{x-6}} \because \frac{d}{dx} [x] = 1 \\ \Rightarrow 1 - (5/\sqrt{c^*-6}) = 0 \because \left. \frac{df}{dx} \right|_{x=c^*} \equiv 1 - (5/\sqrt{c^*-6}) \\ \Rightarrow -5/\sqrt{c^*-6} = -1 \because c + f(x) = -c \\ \Rightarrow -\frac{1}{\cancel{\sqrt{c^*-6}}} \cdot \frac{\cancel{\sqrt{c^*-6}}}{\sqrt{c^*-6}} = -\frac{1}{5} \because \text{Transposition, } \frac{1}{n} \cdot \frac{n}{d} = \frac{1}{n} f(x) \\ \Rightarrow 5 = \sqrt{6-c^*} \because \frac{n_1}{d_1} \cdot \frac{n_2}{d_2} = n_1 \cdot d_2 = d_1 \cdot n_2 \\ \Rightarrow c^* = -19 \because \text{Basic Algebra (somewhat condensed).} \end{array} \right.$$

With this method, I have worked out that the optimal value of the given function is $c^* = -19$.

Step 3: Function Behaviour

The main task at hand is to discover the intervals in which the function's instantaneous rate of change is increasing, and when it is decreasing. My particular solution to this problem somewhat lacks finesse, but it does "work" for this particular purpose. The method is to construct a table of $f(x)$ values within a particular range within some $c^* \pm n \cdot \Delta x$ band. For this case, I have decided on $\Delta x = 2$ and to compute on a range of $[-4 \leq n \leq 4]$:

n	$x = c^* + n \cdot \Delta x$ Specifically, $x = -19 + n \cdot 2$	$f(x)$
-4	-27	30.45
-3	-25	30.68
-2	-23	30.85
-1	-21	30.96
0	-19	31.00
1	-17	30.95
2	-15	30.82
3	-13	30.58
4	-11	30.23

From a value of n starting at -23, and ending at -19, the function output is slowly increasing. Now, from an n value from -19 to -15, the value is decreasing. From this cursory analysis, I have concluded that the increasing interval ($I_{\Delta f(x)} > 0$) and the decreasing interval ($I_{\Delta f(x)} < 0$) are worked out to:

$$I_{[\Delta f(x) > 0]} = (-\infty, -19)$$

$$I_{[\Delta f(x) < 0]} = (-19, +\infty)$$

Ungraded Grade: 0/0.0

✖ Total grade: $1.0 \times 1/2 + 0.0 \times 1/2 + 0.0 \times 0/2 = 50\% + 0\% + 0\%$

Question2: Score 0/1

The First Gate

Having crossed the room, you find yourself at another gate with a required passcode. "Computer, what do I need to enter for this gate to open?" Your computer responds with the following:

Find the absolute maximum and absolute minimum of the function $f(x) = 5x^2 - 10x + 3$ on the interval $[-3, 2]$. These values will be used to generate the passcode.

Absolute Maximum	Absolute Minimum

Your response	Correct response
-3	78
Auto graded	Grade: 0/1.0 

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Your response	Correct response
1	-2
Auto graded	Grade: 0/1.0 

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Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

The following function is given:

$$f(x) = \left\{ -3 \leq x \leq 2 : 5x^2 - 10x + 3 \right\}$$

The task is to work out an $x_{\min}$ and an $x_{\max}$ such that:

$$\begin{aligned} x_{\min} &= c; \text{ s.t. } (\forall x \in [-3, 2]) : f(x) \geq f(c) \\ x_{\max} &= c; \text{ s.t. } (\forall x \in [-3, 2]) : f(x) \leq f(c) \end{aligned}$$

I began by working out the value of the vertex, which is the $x_{\min}$ of the curve, which I accomplished by taking its derivative:

$$\begin{aligned} \operatorname{argmin}_x \longrightarrow \text{sol.} \left\{ \begin{aligned} &\frac{df}{dx} \Big|_{x=c*} = 0 \because \text{Absolute Minimum Problem Setup} \\ &\Rightarrow \frac{d}{dx} [5x^2 - 10x + 3] \\ &\because f(x) = 5x^2 - 10x + 3 \\ &= \frac{d}{dx} [5x^2] - \frac{d}{dx} [10x] + \frac{d}{dx} [3] \because \frac{d}{dx} \sum f(x) = \sum \frac{d}{dx} f(x) \\ &= 5 \cdot \frac{d}{dx} [x^2] - 10 \cdot \frac{d}{dx} [x] + \frac{d}{dx} [3] \because \frac{d}{dx} [k \cdot f(x)] \\ &= 5 \cdot [2 \cdot x^{2-1}] - 10 \cdot \frac{d}{dx} [x] + \frac{d}{dx} [3] \because \frac{d}{dx} [x^n] = n \cdot x^{n-1} \\ &= 5 \cdot [2 \cdot x^{2-1}] - 10 \cdot [1] + \frac{d}{dx} [3] \because \frac{d}{dx} [x] = 1 \\ &= 5 \cdot [2 \cdot x^{2-1}] - 10 \cdot [1] + 0 \because \frac{d}{dx} [c] = 0 \\ &= 10x - 10 \because \text{Simplifications} \\ &\Rightarrow 10c* - 10 = 0 \\ &\because \text{The problem setup is: } \frac{df}{dx} \Big|_{x=c*} = 0 \\ &\Rightarrow 10c* = 0 + 10 \\ &\because \text{Transposition: } ax + b = 0 \Rightarrow ax = -b \\ &\Rightarrow c* = 10 \div 10 = 1 \\ &\because \text{Further Transposition: } ax = -b \Rightarrow \frac{1}{a} \cdot \frac{ax}{1} \end{aligned} \right. \end{aligned}$$

With this logic, I have worked out the solution to $x_{\min}$ to be $x_{\min} = 1$.

Next, I need to work out the absolute maxima, or the $x_{\max}$. The solution to this kind of problem is more intuitive: the quadratic function grows in the positive direction of the ordinate. The function is bounded with a lower bound of -3, and an upper bound of 2. The solution to the $x_{\max}$ absolute maxima is to compute the $f(x)$ where x is both -3, the lower bound, and 2, or the upper bound, and then select the greater value:

$$\begin{aligned} x_{\max} &= \operatorname{argmax}_{x \in [-3, 2]} f(x) \\ \text{Function Evaluations} &\longrightarrow \begin{cases} f(\text{lower bound}) = f(-3) = +78 \\ f(\text{upper bound}) = f(+2) = +3 \end{cases} \end{aligned}$$

Clearly, the lower bound gives the greater value. From this, I can conclude that the absolute minimum is at $x_{\min} = 1$ and that the absolute maximum is at $x_{\max} = -3$.

Ungraded Grade: 0/0.0

✖ Total grade: 0.0×1/2 + 0.0×1/2 + 0.0×0/2 = 0% + 0% + 0%

Question3: Score 1/1

Growing the Raw Materials

As the gate opens, you step into a room that appears to be a laboratory of some kind. Your computer informs you that this is where the raw materials that power the spacecraft are grown. "What do you mean, grown?", you ask. Your computer explains that a special bacteria is grown in this laboratory and converted into fuel. A specific amount of bacteria must be present for the next step in the process to work (within a reasonable margin of error). The computer provides you with information about the population of the bacteria, which follow an exponential model, shown below. In this model, P is the population of bacteria (total count, measured in millions) and t is the time measured in minutes. The population will grow based on this model once a light source is provided and stop immediately once the light source is removed.

$$P(t) = 7e^{t/8}$$

Your computer continues with a note from the files that the alien civilization performed these calculations on the linearization of $P(t)$. Therefore, you will need to linearize $P(t)$ and then use that model to determine when to remove the light source to have 7.238 million bacteria. Your linearization should be around $t = 0$, since this is the time when the light source is switched on.

$$L(t) =$$

Your response	Correct response
7+.875*t	7+0.125*7*t

Auto graded Grade: 1/1.0

Using this linearization, after how many minutes (to three decimal places) will you turn off the light to generate a population of 7.238 million bacteria?

$$t =$$

Your response	Correct response
0.272	0.272±0.001

Auto graded Grade: 1/1.0

minutes

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

The given problem is to grow bacteria by placing light onto a sample of said bacteria. The growth rate of the bacteria is dependent on the amount of time that it is exposed to the light source, as expressed by the following model:

$$P(t) = 7e^{t/8}$$

This experiment has been performed before, with the demonstrators using linearization to estimate the growth rate, suggesting that in order for us to totally reproduce their work, we should follow their procedure as closely as possible. Linearization takes the following form:

$$\text{General Notion} \longrightarrow L(x = a) = f(a) + \left. \frac{df}{dx} \right|_{x=a} \cdot (x - a)$$

where the a coefficient ought to be as close to zero (0) as possible. So, specifically, linearization is applied to this problem and set up as:

$$\text{Specific Case} \longrightarrow L_P(t = 0) = P(0) + \left. \frac{dP}{dt} \right|_{t=0} \cdot (t - 0)$$

The solution involves evaluating the $P(t)$ at $t = 0$, then evaluating $\left. \frac{dP}{dt} \right|_{t=0}$, and finally substituting the worked $P(0)$ and $\left. \frac{dP}{dt} \right|_{t=0}$ into the $L_P(t)$ and simplifying it.

Working out the constituent $P(0)$ is fairly trivial:

$$P(0) = 7e^{0/8} = 7$$

The somewhat more tedious part is to work out the $\left. \frac{dP}{dt} \right|_{t=0}$, which is depicted by the following method below:

$$\left. \frac{dP}{dt} \right|_{t=0} \xrightarrow{\text{sol.}} \begin{cases} \left. \frac{dP}{dt} \right|_{t=0} \because \text{Given} \\ = \left. \frac{d}{dt} \right|_{t=0} [7e^{t/8}] \because P(t) = 7e^{t/8} \\ = 7 \cdot \left. \frac{d}{dt} \right|_{t=0} [e^{t/8}] \because \frac{d}{dt} [c \cdot P(t)] = c \cdot \frac{d}{dt} [P(t)] \\ = 7 \cdot \left[\frac{e^{t/8}}{8} \right]_{t=8} \because \frac{d}{dt} [e^{P(t)}] = \frac{d[e^t]}{d[t/8]} \cdot \frac{d[t/8]}{dt} \\ = 0.875 \because \text{Evaluation and Simplification} \end{cases}$$

With the constituent $P(0)$ and $\left. \frac{dP}{dt} \right|_{t=0}$ values worked out, the $L_P(t)$ estimation can be constructed:

$$L_P(t) = P(0) + \left. \frac{dP}{dt} \right|_{t=0} \cdot (t - 0)$$

$$\text{Worked Solution} \longrightarrow L_P(t) = 7 + 0.875t$$

With this linearized estimator worked out for a value such that $P(t = 0)$, I can now proceed to estimate the number of minutes needed to expose the bacteria sample to a light source in order for it to grow to a target population of 7.238 million individual bacteria. The units for the formulae are t [minutes] and $P(t)$ $\left[\frac{\text{million bacteria}}{\text{per minute (t)}} \right]$, and the linearization formula is set up as:

$$7.238 = 7 + 0.875t$$

The solution is to solve for t with basic algebra:

$$\begin{aligned} 7.238 &= 7 + 0.875t \\ \Rightarrow 0.875t &= 7.238 - 7.000 = 0.238 \\ \Rightarrow \frac{0.875t}{0.875} &= \frac{0.238}{0.875} \\ \Rightarrow t &= 0.272 \text{ [minutes]} \end{aligned}$$

Thus, it would take about 0.272 minutes, or about 16.3 seconds, to grow the bacteria colony up to about 7.238 million individual units of bacteria.

Ungraded **Grade:** 0/0.0



Total grade: $1.0 \times 1/2 + 1.0 \times 1/2 + 0.0 \times 0/2 = 50\% + 50\% + 0\%$

Question4: *Score 0/1*

Creating the Fuel

You now have the bacteria needed to create the fuel. "Computer, what is the next step in this process?" Your computer responds that the bacteria have given off enough gas (collected in a 17 L closed container) to create the fuel, and that the fuel is developed by increasing the temperature of the gas at a rate such that the pressure will initially rise at a rate of 24.9435 kPa/min (kilopascals per minute). You control the initial rate of change of temperature, and must set this value so that the initial rate of change of pressure is 24.9435 kPa/min. Once you have made this initial adjustment, the system will adjust dynamically until the fuel is created.

Your computer informs you that this gas follows the Ideal Gas Law ($PV = nRT$), and that the units for these calculations are already accounted for. This means that no conversions are necessary in your calculations. Some initial values (at time $t = 0$) are provided below:

Pressure (P) = 102 kPa

Volume (V) = 17 L

nR (a constant) = 8.3145 kPaL/K

Temperature (T) = 311 K

What value do you provide to your computer for the initial rate of change of temperature in kelvin/minute?

Rate of Change of Temperature

Your response		Correct response	
0.01438		51±0.001	
Auto graded	Grade: 0/1.0 		

K/min

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

Note that: I did not have enough time to finish up this problem. The following are a part of my notes when attempting this problem:

This is a related rates problem that applies the notion of the ideal gas law to predict the pressure in a gas tank [1], which posits that:

$$PV = nRT$$

Here, P is the amount of pressure experienced in the chamber, V is the volume of the chamber, nR is a constant, and T is the temperature inside the chamber. For this hypothetical problem, the exact units of the features in this model are irrelevant (as the information system will automatically do conversions), and what is of importance is to set the rate of change of the T to an appropriate value such that the product of the pressure and volume of the chamber, its PV , rises to

$PV = 24.9435 \text{ [kPa/min]}$, with the following values for the ideal gas law set as:

$$\text{Initial Conditions} \rightarrow \begin{cases} P &= 102 \text{ [kPa]} \\ V &= 17 \text{ [L]} \\ nR &= 8.3145 \text{ [kPaL/K]} \\ T &= 311 \text{ [K]} \end{cases}$$

The procedure, that I have liberally adopted from the kind student Nathan Pflueger [1], that I have used to work out this solution is as follows: take the natural logarithm of the features in the ideal gas law model, set them up as a related rate problem, and then work out the ideal, initial T value. I started by taking the derivatives of the log of the ideal gas law, and then implicitly differentiating P with respect to T :

$$\begin{aligned} PV &= nRT \\ \Rightarrow PV &= T \because nR = 0 \\ \Rightarrow \ln(PV) &= \ln(T) \because \text{Convenience} \\ \Rightarrow \ln(P) + \ln(V) &= \ln(T) \because \ln(xy) = \ln(x) + \ln(y) \\ \Rightarrow \frac{dP}{dT} \cdot \ln(P) + \ln(V) \cdot \frac{dP}{dT} &= \frac{dT}{dT} [\ln(T)] \\ \because \frac{d}{dx} [\ln(x)] &= \frac{1}{x} \wedge \text{Setup for Implicit Differentiation} \\ \Rightarrow \frac{1}{P} \cdot \frac{dP}{dT} + \frac{1}{V} \cdot \frac{dP}{dT} &= \frac{1}{T} \\ \Rightarrow \frac{dP}{dT} \left(\frac{1}{P} + \frac{1}{V} \right) &= \frac{1}{T} \because \text{Factoring} \\ \Rightarrow \frac{dP}{dT} \frac{\left(\frac{1}{P} + \frac{1}{V} \right)}{\left(\frac{1}{P} + \frac{1}{V} \right)} &= \frac{\frac{1}{T}}{\frac{1}{P} + \frac{1}{V}} = \frac{PV}{T} \\ \because (a+b)x &= c \Rightarrow x = \frac{c}{a+b} \wedge \frac{\frac{1}{a}}{\frac{1}{b} + \frac{1}{c}} = \frac{bc}{a} \end{aligned}$$

From this assessment, I have worked out that the change in pressure, with respect to temperature (which can be changed) is equal to the product of the total pressure and the volume of the chamber, divided by the current temperature of the system. With this model,

Endnotes

1. I would like to acknowledge Nathan Pflueger (2013) for their lecture notes on related rates, which included an example of applying related rates and the ideal gas law to this toy engineering problem (retrieved on Dec. 15, 2025): <https://npflueger.github.io/teaching/math1a/lecture22.pdf>

Ungraded Grade: 0/0.0



Total grade: 0.0×1/1 + 0.0×0/1 = 0% + 0%

Question5: Score 0/1

Constructing the Fuel Tank

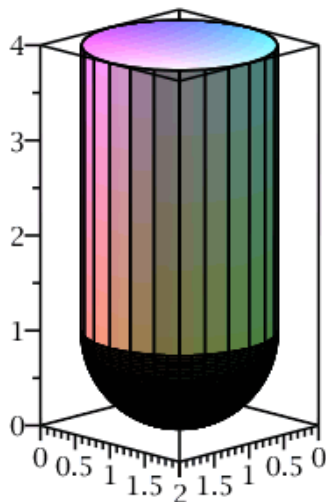
"Great, I've got the fuel, but how do I get it to a spacecraft? Computer, how do I transport this fuel?" Your computer responds that the burning of the fuel destroys the container that it is held in, and that a new container is constructed for each launch of the spacecraft. The shape of the container is a cylinder with a hemisphere attached to the bottom. Your computer is able to generate an image (not to scale) and equations for the volume and surface area of this shape.

You see a warning notice that supplies of materials for constructing this container are low. In order to create the container with limited supplies, you will need to provide the height and radius (to 2 decimal places) that minimizes the surface area of the container. What values do you provide?

Total Volume of Fuel: 11π

Equation for Volume of this Container: $V = \pi r^2 \left(h + \frac{2}{3}r \right)$

Equation for the Surface Area of this Container: $S = 2\pi r h + 3\pi r^2$



Radius:

Your response	Correct response
	1.875777±0.01

Auto graded Grade: 0/1.0 ✖

Height:

Your response	Correct response
	1.875777±0.01

Auto graded Grade: 0/1.0 ✖

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

Ungraded Grade: 0/0.0

✖ Total grade: 0.0×1/2 + 0.0×1/2 + 0.0×0/2 = 0% + 0% + 0%

Question6: Score 0/1

Mixing the Fuel

Having finalized the container, you are notified by your computer that you need to mix a stabilizer with the fuel to ensure that it survives the time to the spacecraft. The computer provides a readout of the current stabilization level along with minimum and maximum levels of stabilization. An equation, $s(t)$, that models the levels of stabilization over time is also provided. You only need to consider these stabilization levels as t gets very large (approaches ∞).

$$s(t) = \frac{\ln(11/t+1)}{16/t}$$

Minimum Level of Stabilization: 0.446875

Maximum Level of Stabilization: 2.7875

If the stabilization level will exceed the maximum allowed level, you must add a destabilizer. If the stabilization level will fall below the minimum allowed level, you need to add more stabilizer. Finally, if the stabilization level is between these bounds, no action is required. What do you do?

Your response	Correct response
Add Stabilizer	Do Nothing

Auto graded Grade: 0/1.0 ✖

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

Unfortunately, I am not quite sure what the question is asking (so I did not spend too much time on it). But my approach was to plot the $s(t)$ and perform a cursory glance on its end behaviour as $t \rightarrow +\infty$.

From my (very cursory) assessment, I worked out that the $s(t)$ never passes the maximum level of stabilization $s(t) > 2.7875$, with the global maximum of $t^* \approx 3.952$ and $s(t^*) \approx 0.197$, and a continuously decreasing slope as $t \rightarrow +\infty$. If the rule is to add more stabilizer as the $s(t) < s_{\min}$ (where $s_{\min} = 0.44685$), then as the function approaches infinity, more stabilizer is needed to keep the fuel tank stabilized.

Ungraded Grade: 0/0.0

✖ Total grade: 0.0×1/1 + 0.0×0/1 = 0% + 0%

Question7: Score 1/1

Charging the Capacitor

Having stabilized the fuel, you look to your computer for next steps. You are told that the spacecraft uses a "start capacitor" that must reach a required voltage in order to start its engine. This voltage is determined by the fixed capacitance and charge as a function of time, $q(t)$. These capacitors are charged and stored in your current location and must be brought to the spacecraft. Your computer provides you with the following definitions:

Charge: The charge as a function of time is $q(t) = \int i(t) dt$,

where $i(t) = 12t^5 + 16t^3 - 6$ is the current as a function of time.

You need to provide the computer with the function $q(t)$ in order for it to appropriately charge the start capacitor. If an unknown constant is needed, it can be included using "+ C", being sure to use a capital letter as the lowercase is used for capacitance.

$q(t) =$

Your response	Correct response
$2*t^6+4*t^4-6*t+C$	$2*t^6+4*t^4+C-6*t$

Auto graded Grade: 3/3.0 ✔

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

The following function is given:

$$i(t) = 12t^5 + 16t^3 - 6$$

The task at hand is to work out its anti-derivative:

$$q(t) = \int i(t) dt$$

I have done this with the following method:

$$\int i(t) dt \xrightarrow{\text{sol.}} \begin{cases} \int i(t) dt \because \text{Given} \\ \Rightarrow \int 12t^5 + 16t^3 - 6 dt \because i(t) = 12t^5 + 16t^3 - 6 \\ \Rightarrow C + \int 12t^5 + 16t^3 - 6 dt \because \int i(t) = C + \int i(t) \\ \Rightarrow C + \int 12t^5 dt + \int 16t^3 dt - \int 6 dt \because C + \int i(t) \\ \Rightarrow C + 12 \cdot \int t^5 dt + 16 \cdot \int t^3 dt - \int 6 dt \because \int \sum i(t) = \sum \int i(t) \\ \Rightarrow C + 12 \cdot \frac{t^{5+1}}{5+1} + 16 \cdot \frac{t^{3+1}}{3+1} - \int 6 dt \because \int t^n = \frac{t^{n+1}}{n+1} \\ \Rightarrow 2t^6 + 4t^4 - 6t + C \because \text{Simplifications} \end{cases}$$

Ungraded Grade: 0/0.0



Total grade: $1.0 \times 3/3 + 0.0 \times 0/3 = 100\% + 0\%$

Question8: Score 1/1

The Second Gate

You arrive at the gate that holds the spaceship. Your computer translates the password requirement as follows:

Find the indefinite integral on the domain of positive real numbers for the following function:

$$f(x) = \frac{15}{x} + \frac{9}{8\sqrt{x}}$$

To appropriately translate the password, you will need to enter an exact answer. Your computer reminds you to include a "+ C" if appropriate, and that you should enclose arguments of functions in parentheses, such as $\sin(2x)$. To enter $\sqrt{a}$, type `sqrt(a)`.

Gate Password:

Your response	Correct response
$15 \cdot \ln(\text{abs}(x)) + (9 \cdot \text{sqrt}(x))/4 + C$	$15 \cdot \ln(\text{abs}(x)) + 9/4 \cdot x^{(1/2)} + C$

Auto graded Grade: 3/3.0

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

The following function is given:

$$f(x) = \frac{15}{x} + \frac{9}{8\sqrt{x}}$$

The task at hand is to work out the antiderivative of the given function, which is formalised as:

$$F(x) = \int f(x) dx$$

I employed the following logic to derive the $F(x)$:

$$F(x) \rightarrow \text{sol.} \left\{ \begin{array}{l} F(x) = \int f(x) dx \because \text{Given} \\ \Rightarrow \int \frac{15}{x} + \frac{9}{8\sqrt{x}} dx \because f(x) = \frac{15}{x} + \frac{9}{8\sqrt{x}} \\ \Rightarrow C + \int \frac{15}{x} + \frac{9}{8\sqrt{x}} dx \because \int f(x) = C + \int f(x) \\ \Rightarrow C + \int \frac{15}{x} dx + \int \frac{9}{8\sqrt{x}} dx \because \int \sum f(x) = \sum \int f(x) \\ \Rightarrow C + 15 \cdot \int \frac{1}{x} dx + \frac{9}{8} \cdot \int \frac{1}{\sqrt{x}} dx \because \int c \cdot f(x) = c \cdot \int f(x) \\ \Rightarrow C + 15 \cdot \ln(|x|) + \frac{9}{8} \cdot \int \frac{1}{\sqrt{x}} dx \because \int \frac{1}{x} = \ln(|x|) \\ \Rightarrow C + 15 \cdot \ln(|x|) + \frac{9}{8} \cdot \int x^{1/2} dx \because \sqrt{x} = x^{1/2} \\ \Rightarrow C + 15 \cdot \ln(|x|) + \frac{9}{8} \cdot \frac{x^{(-1/2)+(2/2)}}{-\frac{1}{2} + \frac{2}{2}} \because \int x^n = \frac{x^{n+1}}{n+1} \\ \Rightarrow 15 \cdot \ln(|x|) + \frac{9\sqrt{x}}{4} + C \because \text{Simplifications} \end{array} \right.$$

With this method, I have concluded that:

$$F(x) = 15 \cdot \ln(|x|) + \frac{9\sqrt{x}}{4} + C$$

Ungraded Grade: 0/0.0



Total grade: $1.0 \times 3/3 + 0.0 \times 0/3 = 100\% + 0\%$

Question9: Score 1/1

The Spacecraft Password

Entering the room, you see the spacecraft. Finally, you will be able to leave this planet! Upon closer inspection, you realize that there is a password to enter the ship. Your computer translates the password as follows:

Find all functions whose derivative is $-7\sin(x)$.

Your computer is able to provide a hint that these functions only differ by a constant and that an arbitrary constant can be included using "+ C". In order to appropriately enter the password, you will need to enclose arguments of functions in parentheses. For example, $\sin(2x)$.

Spacecraft Password:

Your response	Correct response
$7*\cos(x)+C$	$7*\cos(x)+C$

Auto graded Grade: 3/3.0 

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

The following expression is given:

$$f(x) = -7 \cdot \sin(x)$$

The task at hand is to work out a general equation $F(x)$ such that $\frac{dF}{dx} = f(x)$. This kind of problem is worked out with the method of integration and antiderivatives, which is formalised as:

$$\text{General Procedure} \longrightarrow \int f(x) dx$$

$$\text{Specifically} \longrightarrow \int -7 \cdot \sin(x) dx$$

Such a solution is trivial: I went about solving it by applying the constant rule, and then working out the antiderivative of the $\cos(x)$, and then simplifying. The following procedure demonstrates what I did:

$$\int -7 \cdot \sin(x) dx \longrightarrow \text{sol.} \left\{ \begin{array}{l} \int -7 \cdot \sin(x) dx \because \text{Given} \\ = C + \int -7 \cdot \sin(x) dx \because \int f(x) = C + \int f(x) \\ = C - 7 \cdot \int \sin(x) dx \because \int c \cdot f(x) = c \cdot \int f(x) \\ = C - 7 \cdot [-\cos(x)] dx \because \int \sin(x) = -\cos(x) \\ = 7 \cdot \cos(x) + C = F(x) \because F(x) = \int f(x) dx \end{array} \right.$$

Of course, one may verify the correctness of the procedure by invoking the fundamental theorem of calculus, which posits that:

$$\frac{dF}{dx} = \frac{d}{dx} \left[\int f(x) dx \right] = f(x)$$

The following argument demonstrates the correctness of my work:

$$\frac{dF}{dx} = f(x) \longrightarrow \text{sol.} \left\{ \begin{array}{l} P_1 : \frac{dF}{dx} = f(x) \because \text{Given} \\ P_2 : \frac{d}{dx} [7 \cdot \cos(x) + C] = -7 \cdot \sin(x) \because \text{Definitions of } F(x) \wedge f(x) \\ P_3 : f(x) = \frac{dF}{dx} = \frac{d}{dx} [7 \cdot \cos(x) + C] \longrightarrow \text{eval.} \\ \quad = \frac{d}{dx} [7 \cdot \cos(x)] + \frac{d}{dx} [C] \because \frac{d}{dx} \sum f(x) = \sum \frac{d}{dx} f(x) \\ \quad = 7 \cdot \frac{d}{dx} [\cos(x)] + \frac{d}{dx} [C] \because \frac{d}{dx} [c \cdot f(x)] = c \cdot \frac{d}{dx} [f(x)] \\ \quad = 7 \cdot [-\sin(x)] + \frac{d}{dx} [C] \because \frac{d}{dx} [\cos(x)] = -\sin(x) \\ \quad = 7 \cdot [-\sin(x)] + 0 \because \frac{d}{dx} [c] = 0 \\ \quad = -7 \cdot \sin(x) \because \text{Simplifications} \\ P_4 : \frac{dF}{dx} = -7 \cdot \sin(x) \because \text{of work done in } P_3 \\ P_5 : f(x) = -7 \cdot \sin(x) \because \text{Declared in } P_2 \\ P_6 : \frac{dF}{dx} = f(x) \because \text{Formal treatment of the evaluation theorem} \\ \longrightarrow \text{Specifically ... } -7 \cdot \sin(x) = -7 \cdot \sin(x) \checkmark \end{array} \right.$$

Thus, with a given set of definitions and rules that are in accordance with the derivative and integral calculi, I have proven that:

$$\int [-7 \sin(x)] dx = 7 \cdot \cos(x) + C$$

where C is an arbitrary constant that is typically interpreted as an initial condition for a specific use of the more general antiderivative.

Ungraded Grade: 0/0.0



Total grade: 1.0×3/3 + 0.0×0/3 = 100% + 0%

Question10: *Score 1/1*

The Computer Password

As you enter the spacecraft, you look to connect your computer. To your dismay, the controls are also password protected. Your computer helpfully translates the prompt and notifies you that three numbers are needed. Your computer is able to determine the algorithm for determining each number, along with some password hints. The computer displays the below information:

Password Hints:

$$\int_{-8}^{10} f(x) \, dx = -1$$

$$\int_{-8}^{10} g(x) \, dx = 16$$

$$\int_{-8}^7 g(x) \, dx = 8$$

First Number: $\int_{-8}^{10} (-4 f(x) + 2 g(x)) \, dx.$

Second Number: $\int_{10}^{-8} (-4 f(x) + 2 g(x)) \, dx + \int_{10}^{10} f(x) \, dx$

Third Number: $\int_7^{10} 2g(x) \, dx$

First Number	Second Number	Third Number

Your response		Correct response
36		36
Auto graded	Grade: 1/1.0 	

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Your response	Correct response
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-36		-36	
Auto graded	Grade: 1/1.0 ✔		

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Your response		Correct response	
16		16	
Auto graded	Grade: 1/1.0 ✔		

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Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

Two functions, $f(x)$ and $g(x)$, are given, but their expressions are not explicitly defined. Their numeric results integrated over some start point x_i and another endpoint x_f are, however, given:

$$\int_{x_i=-8}^{x_f=10} f(x) dx = -1$$

$$\int_{x_i=-8}^{x_f=10} g(x) dx = 16$$

$$\int_{x_i=-8}^{x_f=10} g(x) dx = 8$$

The task-at-hand is to work out three "password numbers" expressed in another set of integrals by applying the rules of integration to the above given formulae. The logic used to work out their results are shown as follows:

The first number is worked out by applying the addition and constant rules of integration onto the expression:

$$\text{First Number} \rightarrow \text{sol.} \left\{ \begin{array}{l} \int_{-8}^{10} [-4f(x) + 2g(x)] dx \because \text{Given} \\ = \int_{-8}^{10} [-4f(x)] dx + \int_{-8}^{10} [2g(x)] dx \\ = -4 \cdot \int_{-8}^{10} [f(x)] dx + 2 \cdot \int_{-8}^{10} [g(x)] dx \\ = -4 \cdot [-1] + 2 \cdot [16] \because \int_{-8}^{10} f(x) = -1 \wedge \int_{-8}^{10} g(x) = 16 \\ = 4 + 32 = 36 \because \text{Arithmetic} \end{array} \right.$$

The second number is worked out in a similar manner as the first number, but given that the x_i and x_f endpoints are flipped on the lower- and upper-bounds of the integral, I applied the notion that $\int_{x_i}^{x_f} g(x) = -\int_{x_f}^{x_i} g(x)$ to the integral before calculating the anti-derivatives of the expression:

$$\text{Second Number} \rightarrow \text{sol.} \left\{ \begin{array}{l} \int_{10}^{-8} [-4f(x) + 2g(x)] dx \because \text{Given} \\ = -\int_{-8}^{10} [-4f(x) + 2g(x)] dx \because \int_{x_f}^{x_i} f(x) = -\int_{x_i}^{x_f} f(x) \\ = -\int_{-8}^{10} [-4f(x)] dx - \int_{-8}^{10} [2g(x)] dx \because \int \sum f(x) = \sum \int f(x) \\ = 4 \cdot \int_{-8}^{10} [f(x)] dx - 2 \cdot \int_{-8}^{10} [g(x)] dx \because \int c \cdot f(x) = c \cdot \int f(x) \\ = 4 \cdot [-1] - 2 \cdot [16] \because \int_{-8}^{10} f(x) = -1 \wedge \int_{-8}^{10} g(x) = 16 \\ = -4 - 32 = -36 \because \text{Arithmetic} \end{array} \right.$$

Finally, the main logic used to work out the third number was to apply the notion that $\int_{x_i}^{x_f} g(x) = -\int_{x_f}^{x_i} g(x)$ to set up the integral in a manner that will allow myself to apply the notion $\int_{x_i}^{x_f} g(x) = \int_{x_i}^c g(x) + \int_c^{x_f} g(x)$ to work it out:

$$\text{Third Number} \longrightarrow \text{sol.} \left\{ \begin{array}{l} P_1 : \int_7^{10} [2 \cdot g(x)] dx \because \text{Given Problem} \\ P_2 : \int_{-8}^7 [2 \cdot g(x)] dx = 8 \because \text{Given Solution} \\ P_3 : \int_{-8}^{10} [2 \cdot g(x)] dx = 16 \because \text{Given Solution} \\ P_4 : \int_{-8}^7 [2 \cdot g(x)] dx = 2 \cdot \int_{-8}^7 [g(x)] dx \because \int c \cdot g(x) \\ \quad = c \cdot \int g(x) \text{ (applied to } P_2) \\ P_5 : 2 \cdot \int_{-8}^7 [g(x)] dx = -2 \cdot \int_7^{-8} [g(x)] dx \\ \quad \because \int_{x_i}^{x_f} [g(x)] = -\int_{x_f}^{x_i} [g(x)] \text{ (applied to } P_4) \\ P_6 : 2 \cdot \left[\int_7^{10} g(x) dx \right] = 2 \cdot \left[-\left[\int_7^{-8} g(x) dx \right] + \left[\int_{-8}^{10} g(x) dx \right] \right] \\ \quad \because \int_{x_i}^{x_f} [g(x)] = \int_{x_i}^c g(x) dx + \int_c^{x_f} [g(x)] dx \\ P_7 : 2 \cdot [-8] + \left[\int_c^{x_f} \right] dx \because \left[-\int_7^{-8} g(x) \right] = -\left[\int_{-8}^7 g(x) \right] = -8 \\ P_8 : 2 \cdot [-8 + 16] \because \int_{-8}^{10} [g(x)] dx = 16 \\ P_9 : 2 \cdot [-8 + 16] = 2 \cdot 8 = 16 \because \text{Arithmetic} \end{array} \right.$$

Ungraded Grade: 0/0.0

✓ Total grade: $1.0 \times 1/3 + 1.0 \times 1/3 + 1.0 \times 1/3 + 0.0 \times 0/3 = 33\% + 33\% + 33\% + 0\%$

Question11: Score 1/1

Plotting the Course

With access to the spacecraft's controls, you connect your computer and begin installing the fuel tank and start capacitor. Your computer informs you that it can plot most of the course to your home planet but needs assistance with determining your velocity and vertical position at a critical moment in time. Your computer provides you with a function that models your acceleration during the first few seconds of flight. You will need to take this information and calculate your velocity and vertical position at time $t = 2$ s. Your computer notes that you can use the below initial conditions in supporting your calculations:

Initial time: 0 s

Initial vertical position: 0 m

Initial velocity: 0 m/s

Acceleration during this time: $a(t) = 68t^3 + 59t$ m/s²

What values (to 2 decimal places) do you provide your computer?

$v(2) =$

Your response	Correct response
390.00	390

Auto graded Grade: 1/1.0 

m/s

$s(2) =$

Your response	Correct response
187.47	187.466667±0.01

Auto graded Grade: 1/1.0 

m

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

The given problem belongs to kinematics. Specifically, a mathematical model is given that models the initial acceleration for "the first few seconds of flight," of an aircraft, and the task at hand is to work out equations of motion regarding the velocity $v(t)$ and displacement $s(t)$ of the aircraft. An non-constant acceleration function is given:

$$a(t) = 68t^3 + 59t \text{ [m/s}^2\text{]}$$

Specifically, I am interested in working out a velocity $v_f = v(2) - v(0) \text{ [m/s]}$ and a vertical displacement

$\Delta s = s(2) - s(0) \text{ [m]}$ where the initial velocity, initial displacement, and initial time are $v_i = 0, s_i = 0$, and $t_i = 0$, respectively.

Velocity is the instantaneous change in displacement, and acceleration is the instantaneous change in velocity. Such an "instantaneous change," is formally treated as a derivative:

$$s(t) = \frac{d}{dt} [v(t)] \text{ [m]}$$

$$v(t) = \frac{d}{dt} [a(t)] \text{ [m/s]}$$

However, the given stuff to work with only involves an acceleration function $a(t)$ and the given initial values. Nonetheless, the problem can still be solved by employing anti-derivatives to work out the expressions for $v(t)$ and $s(t)$ which can then be used to predict their respective velocity and vertical displacement at an arbitrary time t . The following is a formal outline the procedure:

$$\begin{aligned} a(t) &\leftarrow \text{Given} \\ \Rightarrow v(t) &= v_i + \int a(t) dt \\ \Rightarrow s(t) &= s_i + \int v(t) dt \end{aligned}$$

After the $v(t)$ and $s(t)$ anti-derivatives have been worked out, I can then proceed to predict the aircraft's velocity and displacement at an arbitrary t value, which for my purposes is set to $t = 2$. The following is a condensed version of how I went about deriving the $v(t)$ and $s(t)$ functions:

$$\begin{aligned} v(t) &= \int a(t) dt = \int [68t^3 + 59t] dt = \frac{68t^4}{4} + \frac{59t^2}{2} + v_i \\ &= 17.0000t^4 + 29.5000t^2 + v_i \leftarrow \text{Worked velocity function "v(t)"} \end{aligned}$$

$$\begin{aligned} s(t) &= \int v(t) dt = \int [17t^4 + 29.5t^2 + v_i] dt = \frac{17t^5}{5} + \frac{29.5t^3}{3} + \frac{v_i t^2}{2} + s_i \\ &= 3.4000t^5 + 9.8333t^3 + \frac{v_i t^2}{2} + s_i \leftarrow \text{Worked displacement function "s(t)"} \end{aligned}$$

As stated earlier, the problem demands that I compute the velocity and displacement at $t = 2$ - with initial time, initial vertical position, and initial velocity set to $t = 0, s_i = 0$, and $v_i = 0$, respectively. The solution, expressed as Δv and Δs , rounded to two decimal numbers, is worked out to:

$$\begin{aligned} \Delta v &= v(2) - v(0) = [17(2)^4 + 29.5(2)^2 + 0] \\ &\quad - [17(0)^4 + 29.5(0)^2 + 0] = 390.0000 - 0 = 390.00 \text{ [m/s]} \\ \Delta s &= s(2) - s(0) = \left[3.4(2)^5 + 9.8333(2)^3 + \frac{0 \cdot (2)^2}{2} + 0 \right] \\ &\quad - \left[3.4(0)^5 + 9.8333(0)^3 + \frac{0 \cdot (0)^2}{2} + 0 \right] = 187.4664 - 0.0000 = 187.47 \text{ [m]} \end{aligned}$$

Note that the velocity and vertical displacement is measured in metres per second [m/s] and metres [m] - respectively, hence the units appended to the result. With this procedure, I have worked out that ...

$$v(2) = 390.00 \text{ [m/s]}$$

and that ...

$$s(2) = 187.47 \text{ [m/s]}$$

Ungraded Grade: 0/0.0

✓ Total grade: $1.0 \times 1/2 + 1.0 \times 1/2 + 0.0 \times 0/2 = 50\% + 50\% + 0\%$

Question12: Score 1/1**Connecting the Start Capacitor**

The spacecraft is nearly ready for takeoff! The last task is to attach the start capacitor to the ship in a simple RC-circuit. Your computer warns you that some adjustments may need to be made if the heat generated in the resistor over the initial time interval is higher than a given threshold.

Your computer is able to provide the following information, along with a note that the units are already accounted for in the equations:

Heat Generated in Resistor over the first 3 seconds: $Q = \frac{V_b^2}{R} \int_0^3 e^{-2t/(RC)} dt$ joules

Voltage of the Battery: $V_b = 2$ volts

Resistance: $R = 2$ ohms

Capacitance: $C = \frac{1}{2}$ farads

Maximum Heat Allowed in Resistor: $Q_{\max} = 1$ joules

Is the heat generated in the resistor over the first 3 seconds below the maximum allowable threshold?

Your response	Correct response
Yes	Yes

Auto graded Grade: 1/1.0 ✓

What is the difference between the maximum threshold (provided above by your computer) and the amount of heat generated in the resistor (that you calculated)? Provide a positive value up to three decimal places.

Your response	Correct response
0.003	0.002479±0.001

Auto graded Grade: 1/1.0 ✓

joules

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

From experience, I recognize this problem as one that concerns itself with heat transfer and steady heat conduction - both of which are a subset of thermodynamics and electrical engineering. The amount of heat generated, or the Q , given the number of time passed in seconds t , amongst other parameters, for this particular electronic device is given as follows:

$$Q(t) = \frac{V_b^2}{R} \int_0^t e^{-2t/(R \cdot C)} dt [J]$$

The parameters in question are the voltage of the battery V_b , its resistance R , and its capacitance $C = \frac{1}{2}$ --- which are parameterized as follows:

$$V_b = 2 [V] \leftarrow \text{measured in volts}$$

$$R = 2 [\Omega] \leftarrow \text{measured in ohms}$$

$$C = \frac{1}{2} [F] \leftarrow \text{measured in faradays}$$

Part 1: Solving the $Q(t)$ equation

Before addressing the task at hand, I will begin by solving the antiderivative of the $e^{-2t/(R \cdot C)}$, which is:

$$\begin{aligned} Q &= \int e^{-2t/(R \cdot C)} dt [J] \\ &= \int e^{-2t/[2 \cdot (1/2)]} dt [J] \\ &\because R = 2, C = (1/2) \\ &= \int e^{-2t/1} dt [J] = \int e^{-2t} dt [J] \\ &= -\frac{1}{2} \cdot e^{-2t} [J] \end{aligned}$$

Since the $Q(t)$ is composed of the $\frac{V_b^2}{R}$ constant, and the previously worked out integral, the general form of the $Q(t)$ is worked out to:

$$\begin{aligned} Q(t) &= \frac{V_b^2}{R} \cdot -\frac{1}{2} \cdot e^{-2t} [J] \\ &= -\frac{V_b^2 \cdot e^{-2t}}{2R} [J] \end{aligned}$$

When substituting $R = 2$ and $V_b = 2$, the specific case of the $Q(t; R = 2, V_b = 2)$ is worked out to:

$$\begin{aligned} Q(t) &= -\frac{V_b^2 \cdot e^{-2t}}{2R} [J] \\ &= -\frac{2^2 \cdot e^{-2t}}{4} [J] = -e^{-2t} [J] \end{aligned}$$

Part 2: The Task at Hand

The specific case of the $Q(t)$ with the given parameters has been worked out to ...

$$Q(t) = -e^{-2t} [J]$$

The two problems are as follows:

1. The electrical device in question will malfunction if the $Q(t) > Q_{\max}$, which in this case is set to $Q_{\max} = 1 [J]$. Determine *whether-or-not* the electronic device will malfunction because the $Q(t) > Q_{\max}$ threshold has been passed.
2. Work out the difference between $Q(t)$ and $Q_{\max}$, which is formalised as $\Delta Q = Q(t) - Q_{\max}$

For this demonstration, I will assume that the $t = 3$; the first problem is to evaluate the following:

$$\begin{aligned} Q(t) &= \int_0^{t=3} Q(t) dt \\ &= \left[-e^{-2t} \right]_{t_i=0}^{t_f=3} = Q(3) - Q(0) \\ &= \left[-e^{-2 \cdot 3} \right] [J] - \left[-e^{-2 \cdot 0} \right] [J] \\ &\approx [-0.002478] - [-1.000000] \approx +0.997[J] \end{aligned}$$

The $Q_{\max} = 1 [J]$, and the electronic device will malfunction if its $Q(3)$ exceeds the $Q_{\max}$ threshold. The $Q(t) \approx 0.997$ technically **does not** exceed the $Q_{\max}$ threshold. The second problem, which is directly related to the first one because it clarifies on the magnitude of the relationship between the $Q(t)$ and the $Q_{\max}$, is to work out the difference between the $Q(3)$ and the $Q_{\max}$,

$$\Delta Q = Q(3) - Q_{\max} \approx 0.003 [J]$$

While the $Q(3)$ *technically* does not exceed the threshold set by $Q_{\max}$, the difference appears to be very small, and a slight change in the V_b, R, C parameters that constitute the $Q(t)$ heat transfer equation, with antecedents that most definitely come from a hostile "outer space" environment, could cause the resulting ΔQ difference to be < 0 , implying that the $Q(t)$ has passed the $Q_{\max}$ threshold. Further investigation may be needed to ensure that the $\Delta Q > 0$.

Ungraded Grade: 0/0.0



Total grade: $1.0 \times 1/2 + 1.0 \times 1/2 + 0.0 \times 0/2 = 50\% + 50\% + 0\%$